

Temporalization of an Atemporal Parent

Morphogenesis, Actuality, Observer Time, and Empirical Identifiability in Tau Core

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Abstract

The Tau Core foundation sequence separates an atemporal morphological parent, physical source realization, observer-relative temporal descent, quantum records and four-dimensional terminal physics. A long subsequent morphogenesis program developed these layers in detail but left its net result distributed across many conditional constructions, no-go theorems and negative data preflights. This paper gives a nonredundant synthesis. It imports, without re-proving, the component results of Foundation Papers I–VII and states one typed composition theorem.

The conservative architecture is not a competition between an atemporal and a temporal ontology. Temporalizable carriers form a subclass of atemporal parent bodies, while a temporal completion is a fibered enhancement carrying a source-owned order structure. On a finite retained relation complex, a prescribed scalar proto-time exists exactly when the occurrence cochain has zero signed period on every cycle after physical-null descent. In the declared enriched actuality completion, a complete quotient-basic occurrence action selects the exact cochain $\omega_{\text{occ}} = -d\mathcal{A}_{\text{occ}}$, and the EOCC response $X_{\text{EOCC}} = -\Gamma^{-1}d\mathcal{A}_{\text{occ}}$ gives $\omega_{\text{occ}}(X_{\text{EOCC}}) > 0$ away from equilibrium. This proves monotone action descent, progressive stabilization and a finite parent-action formation budget inside that completion. It does not derive EOCC from the static minimal theory or prove that Nature occupies the enriched model.

Observer time is a later body-conditioned, source- and observer-dependent readout. Stable quantizer cells, record instruments and calibration are required before proto-order becomes an operational clock. A frozen past is therefore persistence of compatible prefix restrictions and stable records, not a parent memory store. Neither EOCC order nor record persistence alone selects an entropy arrow. Entropy growth follows conditionally from a source-owned non-erasing visible refinement with one consistent ensemble; thermodynamic promotion additionally requires a physical macroterminal, conserved constraints and an equilibrium measure. Common ownership of the actuality and readout packet does not fix the macroterminal sign. A minimal free-action completion proves local thermodynamic growth only after a positive temperature-like scale and conservation law are supplied. More sharply, a symmetric configuration Laplacian has an exact logarithmic-mean EOCC representation and free-action decomposition. This thermal production need not equal the entropy of a refined front readout. Every finite source-frozen acyclic process in the declared intervention class has an operationally identical atemporal simultaneous factorization; unrestricted finite internal data cannot identify literal ontic traversal. A lawful test must instead freeze a bounded atemporal comparator and detect a source-owned nonfactorizing rank, capacity or record effect. Existing public-data preflights do not yet supply such a signal. Finally, a conditional four-dimensional exchange current can carry the finite formation budget, but its units, receiver equation of state, localization and dark-sector split are not selected. The current status is a mathematically specified conditional temporal working completion with exact internal theorems and exact observational ceilings, not an empirically supported theory of parent time.

Nonredundancy and ownership. Foundation Paper I owns the atemporal parent/readout architecture; Paper II the typed source-signature constraints; Paper III record transport; Paper IV the parent-law realization and actuality boundary; Paper V temporal descent, exact finite temporalizability and the interventional simulation no-go; Paper VI quantizer, complete-record and history-refinement results; Paper VII the conditional 4D terminal boundary. This paper does not reproduce those derivations. It owns their typed composition, the consolidated current-status ledger and the resulting proof/falsification ladder.

Claim boundary. Nothing below proves that old-bare Tau entails an actuality law, that Nature selects the enriched temporal completion, that an absolute parent clock exists, or that observer time equals proto-time. No SI clock or energy unit, dark component, cosmological anomaly, quantum anomaly, or Tau Core signal is derived from the temporal construction. Conditional mathematical closure is kept separate from physical source selection and empirical validation.

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1 Question, method and relation to the foundation sequence

The research question is deliberately narrower than “does time exist in the Parent?” It is

Under which source-owned conditions can an atemporal parent morphology carry an integrable, physically actual temporalization whose observer descent has operational consequences, and which observations could distinguish that completion from a controlled atemporal comparator?

The foundation sequence already fixes the direction

$$B_{\tau}^{\text{phys}} + s_U \longrightarrow M_{\tau} \longrightarrow \mathcal{C}_{OS}(M_{\tau}) \longrightarrow \mathcal{R}_{OS} \longrightarrow Y_{OS}^{(a)}, \quad (1)$$

where s_U is one universal morphology seed, M_{τ} is the stabilized morphological body, $\mathcal{C}_{OS}$ is the body-conditioned full observer–source channel, $\mathcal{R}_{OS}$ is a shared resolved descriptor and $Y_{OS}^{(a)}$ denotes typed terminal readouts. The body is not constructed backward from a clock, quantum outcome, residual or cosmological fit [1–3].

Paper IV isolates the physical source-realization question and the additional actuality content required by a temporal completion [4]. Paper V derives temporal descent and observer clocks from an already available body packet, not the body from time [5]. Papers VI and VII then constrain record and 4D terminal interpretations [6, 7]. This manuscript composes those arrows once. It does not make each terminal paper repeat the parent architecture.

The method has four rules.

- (M1) Every imported result retains its canonical owner and premise set.
- (M2) A current net verdict replaces superseded intermediate gates; negative paths remain provenance, not additional positive claims.
- (M3) Every positive completion is accompanied by an identical-reduct or deletion control showing which additional premise carries the result.
- (M4) Internal mathematical closure, Nature-level source selection and empirical discrimination are reported at different levels.

Figure 1 summarizes the ownership map. The full compact ledger is included in the reproducibility package.

2 Inherited-results ledger

Table 1 is the paper-facing dependency ledger. “Exact” always means exact on the declared finite or regular domain, not physical selection by Nature.

Table 1: Inherited results, owners and use in this synthesis.

Owner	Imported object	Status retained here	Use; not repeated here
Paper I	Atemporal body and forward parent-to-readout direction	Architecture plus conditional MVP	Fixes Eq. (1)

Owner	Imported object	Status retained here	Use; not repeated here
Paper II	Typed source signatures and joint incidence	Finite exact/conditional representation results; one joint $\mathfrak{J}_* = (\Pi_*, \varphi_*)$ fixes mixed ideal, separator multiplicity, relative gluing and faithful block occupation	Prevents untyped source identification and independent retuning of those four descendants
Paper III	Complete record edges and recovery-correctable transport	Exact and conditional record results	Supplies stable record transport without parent history storage
Paper IV	Parent-law realization and EOCC boundary	Conditional completion plus non-entailment	Separates stationary body realization from physical actuality
Paper V	Temporalization fiber, order, clocks and finite process no-go	Exact finite theorems under declared assumptions	Supplies the temporal descent layer
Paper VI	Observer quantization, complete record jets and no-retro refinement	Conditional quantum/readout theorems	Supplies the operational record layer
Paper VII	Gravity, matter and gauge assembly	Conditional terminal recovery only	Blocks direct identification of actuality with 4D stress-energy
TMB 379–391	Seed key, body realization and internal-observer ontology	Typed ontology decision	Fixes the meaning of seed, body and observer
TMB 402–424	Lossy relay, stabilization, EOCC working completion and budget	Conditional theorem family with exact controls	Supplies progressive formation and resolution-relative freezing
TMB 425–440	Formation exchange and dark-sector forks	Conditional construction plus non-entailment	Fixes the 4D energy boundary
TMB 407–413, 442–550	Comparator capacity and observational ceiling	No-go plus bounded-discriminator criteria	Supplies the proof ladder
TMB 551–555	Record-cell window, Choi detection and history refinement	Exact/conditional readout results	Connects continuous formation to records

The compression recorded in the Paper-II row is not a Nature-level selection theorem. Same proper restrictions and local two-jets can support inequivalent mixed cross moments, so the current narrow source does not entail $\mathfrak{J}_*$. The temporal working completion may inherit this one joint datum, but neither temporal order nor terminal agreement can be used backward to select it. Accordingly, the former separator subquestions are closed as independent internal gates, while parent-law realization and global terminal completion remain one irreducible physical boundary.

This ownership split is substantive. For example, Paper V already proves that order does not determine duration, and Paper VI already proves that stable record invariance alone does not imply no-signalling. Restating their proofs would add length but no knowledge. The new task is to determine what follows when their outputs are placed on the same typed source chain.

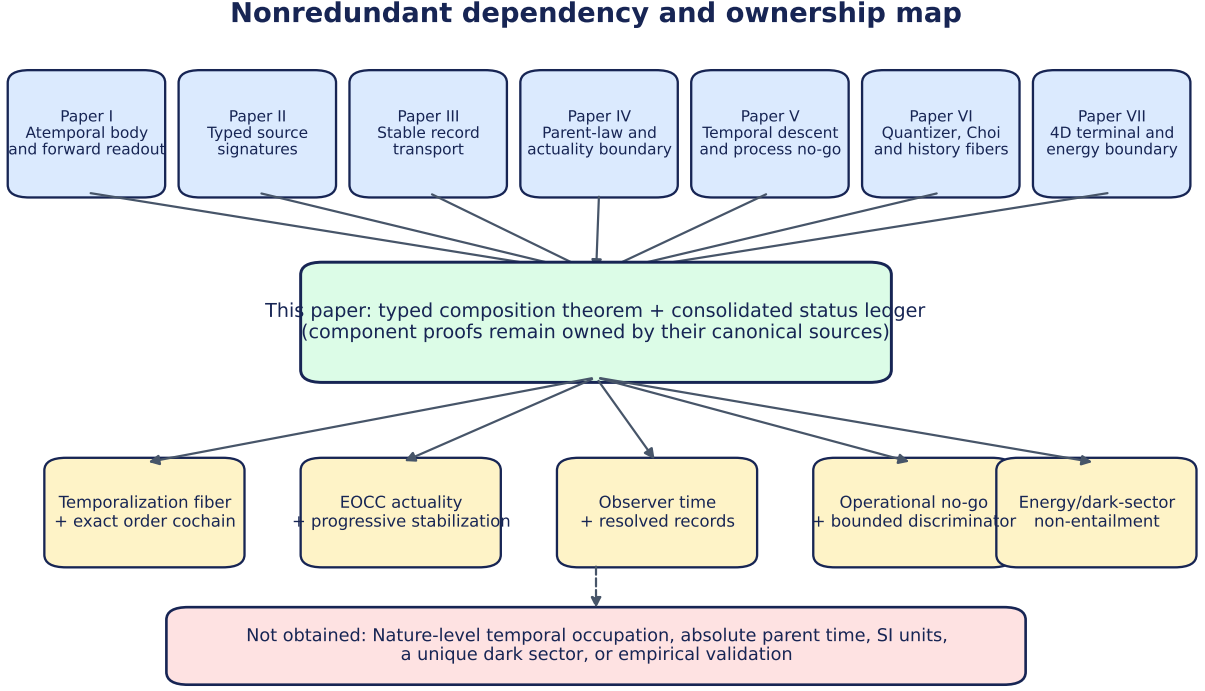


Figure 1: Nonredundant dependency map. Blue boxes are inherited foundation results, the green box is the composition owned here, yellow boxes are conditional consequences, and the red box records conclusions that do not follow. Arrows denote logical dependence, not parent-time evolution.

3 Typed ontology: carrier, temporalization, realization and observer

3.1 The seed is a generative key, not a miniature body

Let $\mathcal{A}_{B,s}(M)$ be one source-frozen base-seed action and let two seeds be response-equivalent when their interaction difference is constant in M . The base-relative morphology key is the equivalence class

$$K_U^{\text{morph}} := [s_U]_B. \quad (2)$$

It supplies a compressed source-side relational asymmetry. It is not an element of the body configuration space. The body is a selected stable response,

$$M_\tau^* = \operatorname{argmin}_{M \in \mathcal{M}_C} \mathcal{A}_{B,s_U}(M), \quad s_U \neq M_\tau^* \text{ by type.} \quad (3)$$

The statement that the seed “contains the universe” is therefore permitted only in the generative sense of Eq. (3), after the base, constraints and response law have been fixed. The same seed reduct may admit stable, unstable or nonforming completions; the stability of our observed universe does not make every admissible seed stable.

3.2 A temporal completion is a fiber, not a replacement ontology

Let $\mathfrak{A}$ be the class of source-admissible atemporal parent bodies after the true physical-null quotient. For $P \in \mathfrak{A}$, let $\text{Temp}(P)$ be the set of source-owned proto-order structures modulo the declared monotone clock gauge. Define

$$\mathfrak{A}_{\text{temp}} = \{P \in \mathfrak{A} : \text{Temp}(P) \neq \emptyset\}, \quad \mathfrak{T} = \bigsqcup_{P \in \mathfrak{A}_{\text{temp}}} \text{Temp}(P) \xrightarrow{\pi} \mathfrak{A}_{\text{temp}} \subseteq \mathfrak{A}. \quad (4)$$

The temporalizable carriers are a subclass of atemporal carriers, whereas the complete temporal theory is generally a fibered enhancement. A single body may admit more than one inequivalent orientation or proto-time chart. Only uniqueness up to clock gauge would make $\mathfrak{T}$ literally identifiable with a subset of $\mathfrak{A}$.

This formulation reconciles two intuitions. The parent body may be described simultaneously, while a selected completion can still possess real ordered actuality. “Atemporal” refers to the carrier ontology; “temporal” refers to additional source-owned structure and its actual occupation. Neither word by itself decides the other’s domain.

3.3 Proto-time and observer time are different typed objects

Proto-time is first an oriented order class on realized source response. It need not have a metric unit. An operational observer is a body-supported relational subsystem whose accessible 4D world and records co-descend from the full channel. Its time is therefore of the form

$$t_O = \text{Cal}_{OS} \circ Q_{OS} \circ \mathsf{T}_{OS}^{\text{clock}} \circ A_{OS}[M_\tau], \quad (5)$$

where A_{OS} is body-conditioned access, $\mathsf{T}_{OS}^{\text{clock}}$ is a typed clock precursor, Q_{OS} is a stable source-derived quantizer and Cal_{OS} supplies a physical clock calibration. None of these terminal maps constructs the seed or body backward.

3.4 Observer ticks do not identify Parent proto-rate

Paper V’s pathwise quantizer theorem sharpens the calibration boundary. On a monotone complete channel path, stable quantizer-cell crossings define an exact tick count invariant under every increasing relabelling of the Parent path parameter. If $\bar{q}$ is the pulled-back common coordinate, $-d\bar{q}/d\rho = \mathcal{T}_q v_r$, and $\nu_O = dN_O/d(-\bar{q}) > 0$ is the oriented observer-cell density, then

$$\boxed{\frac{dN_O}{d\rho} = \nu_O(\bar{q}) \mathcal{T}_q(\bar{q}) v_r(\bar{q})}. \quad (6)$$

The exact reciprocal counterfamily $v_r \mapsto cv_r$, $\nu_O \mapsto \nu_O/c$, for any positive c , preserves the observed tick intensity. Unknown channel transmission adds a third degeneracy. Consequently the amount or intensity of observer change contains proto-rate information but does not identify Parent front rate without independently source-frozen quantizer and transmission calibration. An SI duration additionally requires a descended unit. Different observer tick counts need not imply different path orientation, and an atemporal ordered path can carry the same boundary-crossing sequence. This is a conditional identifiability theorem, not evidence for literal traversal.

The conclusion is unchanged for arbitrarily many observers. If they share one path and common transmission, then

$$\frac{I_i}{I_j} = \frac{\nu_i}{\nu_j}, \quad \text{rank} [\text{Id}_m \quad \mathbf{1} \quad \mathbf{1}] = m, \quad \dim \ker [\text{Id}_m \quad \mathbf{1} \quad \mathbf{1}] = 2. \quad (7)$$

Thus their ratios determine only relative cell densities. One independently known absolute density identifies $\mathcal{T}_q v_r$; an independent transmission identifies v_r only in the chosen Parent parameter. Positive reparameterization preserves every integrated count while changing that numerical rate, so an absolute proto-rate also needs a Parent scale or reference process. Terminal SI clocks cannot be lifted backward to supply that source datum.

3.4.1 Source-calibrated rate and the remaining ontological stop

The non-identifiability theorem does not forbid a positive completion. If one preterminal source packet jointly owns the registered density ν_O , the transmission $\mathcal{T}_q$, and a positive temporalizing scale one-form

$$d\tau_P = \Theta_P = \kappa_P(\rho) d\rho, \quad \kappa_P > 0, \quad (8)$$

then

$$\boxed{u_r = -\frac{dr}{d\tau_P} = \frac{I_O}{\nu_O \mathcal{T}_q \kappa_P}} \quad (9)$$

is invariant under every increasing path reparameterization. Indeed, I_O , v_r , and κ_P all acquire the same inverse Jacobian. For the explicit WR-T60 half-transmission candidate with $\nu_O = 1/\Delta\bar{q}_O$, this gives

$$u_r = \frac{2\Delta\bar{q}_O}{T_*} \frac{dN_O}{d\rho}. \quad (10)$$

This is a mathematically nonempty calibrated temporal completion, not a Nature-level selection theorem. The current source does not jointly own its registered density, transmission, and scale form. Moreover, $\Theta_P \mapsto e\Theta_P$, $e > 0$, preserves the complete tick record while sending $u_r \mapsto u_r/e$. An atemporal ordered carrier can carry the same calibrated measure and boundary sequence. Thus calibrated order does not by itself prove successive actualization.

The rate-dependent formation law is not a new missing object: it is already present conditionally in the enriched EOCC packet. Write it in the path coordinate as

$$\frac{dq}{d\rho} = -\Gamma_\rho^{-1} d\mathcal{A}_{\text{occ}}. \quad (11)$$

For $\lambda = f(\rho)$, the tangent covariance requires $\Gamma_\lambda = f'\Gamma_\rho$, while the clock density obeys $\kappa_\lambda = \kappa_\rho/f'$. Therefore

$$\boxed{\hat{\Gamma} = \kappa_P \Gamma_\rho}, \quad \boxed{\frac{dq}{d\tau_P} = -\hat{\Gamma}^{-1} d\mathcal{A}_{\text{occ}}} \quad (12)$$

are invariant. The physical action-release certificate is

$$-\frac{d\mathcal{A}_{\text{occ}}}{d\tau_P} = \left\langle d\mathcal{A}_{\text{occ}}, \hat{\Gamma}^{-1} d\mathcal{A}_{\text{occ}} \right\rangle \geq 0, \quad (13)$$

with equality at stabilization. This closes the clock–mobility bridge conditionally on joint source ownership; it supplies neither an SI unit nor Nature-level occupation.

The stronger ontology claim remains blocked exactly. The complete autonomous flow has the simultaneous calibrated orbit relation

$$\mathcal{O}_{q_0}^{\text{cal}} = \{[\tau_P, q] \mid q = \Phi_{\tau_P} q_0\}, \quad \Phi_{\tau+h} = \Phi_h \circ \Phi_\tau, \quad (14)$$

which reproduces the same states, currents, release ledger, ticks and finite source-frozen interventions as successive EOCC occupation. More generally, let $\Sigma_{\text{cal}}(\mathcal{T}_{\text{cal}})$ retain simultaneously the complete calibrated orbit, observer–source packet and every admitted terminal map. Then

$$p_{\mathcal{T}_{\text{cal}}}(y \mid e) = p_{\Sigma_{\text{cal}}(\mathcal{T}_{\text{cal}})}(y \mid e), \quad \boxed{B_{\mathcal{T}_{\text{cal}}, \Sigma_{\text{cal}}}(D) = 1} \quad (15)$$

for every admitted protocol, record and finite dataset. The Bayes-factor statement is pairwise; model-family evidence also depends on parameter maps and priors. Thus EOCC actuality remains a primitive clause of the enriched temporal completion. A nonfactorizing terminal can reject a predeclared orbit-only or otherwise bounded atemporal class, but saturation retains that terminal as a simultaneous typed relation. Excluding saturation requires an independently justified source-admissibility law. This possibility is itself limited: if $\mathcal{L}_{\text{ext}}$ contains the complete shared typed source, orbit, observer and terminal packet, then the two reducts are identical and every isomorphism-invariant source condition obeys

$$\mathcal{T}_{\text{cal}}|_{\mathcal{L}_{\text{ext}}} \models \varphi \iff \Sigma_{\text{cal}}(\mathcal{T}_{\text{cal}})|_{\mathcal{L}_{\text{ext}}} \models \varphi. \quad (16)$$

Every uniquely definable enrichment remains tied as well. An extensional occurrence predicate is copyable; giving it irreducibly successive semantics adopts a theory-internal temporal ontology

rather than deriving Nature-level traversal. The remaining noncircular empirical route is a source-derived physical resource bound defining a bounded atemporal comparator. That route is not an absolute separator when the resource is itself a functional of the shared reduct:

$$\mathcal{C}_{\text{ext}}(\mathcal{T}_{\text{cal}}|_{\mathcal{L}_{\text{ext}}}) = \mathcal{C}_{\text{ext}}(\Sigma_{\text{cal}}(\mathcal{T}_{\text{cal}})|_{\mathcal{L}_{\text{ext}}}). \quad (17)$$

If a complete operational cut factors through V_k , the two paired response tables are equal and both obey $\text{rank } K^{(k)} \leq \dim V_k$. A finite sequential realization also has a simultaneous constraint-network dual with the same boundary-state bond dimension. Therefore capacity can reject a preregistered bounded operational class, but cannot reject only its atemporal interpretation.

This separates Tau temporalization from stationary-observable accounts of emergent time and from causal-order approaches while retaining a family resemblance: correlations or order may be primary even when an external master clock is absent [8–11]. The Tau-specific content is the morphology/source/readout typing, not the generic statement that time might be relational.

4 Temporalization synthesis theorem

We now state the one result owned by this manuscript. Its premises are assembled from the canonical owners in Table 1; no premise is silently upgraded from a candidate to a fact about Nature.

Definition 1 (Temporal working packet). *A temporal working packet is*

$$\mathfrak{W}_T = (P, Q_{\text{occ}}, \mathcal{A}_{\text{occ}}, \Gamma, X_{\text{EOCC}}, \mathcal{F}, \mathcal{I}_{OS}, Q_{OS}, \text{Cal}_{OS}), \quad (18)$$

with the following clauses.

- (S1) $P \in \mathfrak{A}_{\text{temp}}$ is an atemporal carrier and Q_{occ} its regular physical occurrence quotient.
- (S2) $\mathcal{A}_{\text{occ}} \in C^1(Q_{\text{occ}})$ is complete, source-frozen and quotient-basic.
- (S3) $\Gamma : TQ_{\text{occ}} \rightarrow T^*Q_{\text{occ}}$ is symmetric positive definite on the retained response domain.
- (S4) EOCC is adopted as the actuality law $X_{\text{EOCC}} = -\Gamma^{-1}d\mathcal{A}_{\text{occ}}$, with regular continuation until stationarity, boundary or loss of regularity.
- (S5) $\mathcal{F} = \{P_a\}$ is a compatible nested prefix/restriction family on the occupied section; it is not a stored trajectory buffer.
- (S6) $\mathcal{I}_{OS}$ is a source-owned complete observer instrument whose stable record partition is Q_{OS} .
- (S7) Cal_{OS} is a separately supplied monotone clock calibration on the resolved ordered records.

Theorem 1 (Temporalization synthesis). *For a temporal working packet satisfying (S1)–(S7):*

- (i) *the temporal completion projects to the atemporal carrier as in Eq. (4);*
- (ii) *the action selects the exact occurrence cochain*

$$\omega_{\text{occ}} := -d\mathcal{A}_{\text{occ}}; \quad (19)$$

- (iii) *on every nonstationary actual section,*

$$\omega_{\text{occ}}(X_{\text{EOCC}}) = \langle d\mathcal{A}_{\text{occ}}, \Gamma^{-1}d\mathcal{A}_{\text{occ}} \rangle > 0, \quad \frac{d\mathcal{A}_{\text{occ}}}{d\sigma} < 0; \quad (20)$$

- (iv) *every finite pullback of ω_{occ} has zero signed cycle period, so it admits a global scalar progress potential unique up to an additive constant on each connected component;*

- (v) *persistent prefixes and stable earlier records are not overwritten by later refinement, but observer time exists only after (S6)–(S7);*
- (vi) *the stabilized endpoint has $d\mathcal{A}_{\text{occ}} = 0$, hence zero formation current and no requirement of continuous maintenance power;*
- (vii) *none of (i)–(vi) makes literal ontic traversal identifiable from every finite internal acyclic intervention protocol.*

Proof. Part (i) is the typed temporalization-fiber result imported from Paper V. By (S2), the differential in Eq. (19) is quotient-basic and exact. Substitution of (S4) gives Eq. (20); positivity of Γ makes the contraction strict away from a stationary point. A finite edge pullback of an exact one-form is an incidence difference, so all signed cycle sums telescope. The finite exact-cochain theorem of Paper V then gives (iv). Clause (S5) and the record-transport results of Paper III give compatible prefix persistence; Paper VI supplies the stable-record and quantizer conditions, while (S7) is required for a clock value. At stationarity the gradient, response and certificate vanish, proving (vi). Finally, the finite interventional simulation theorem of Paper V constructs an atemporal simultaneous factorization with the same terminal and counterfactual tables for every process in the declared finite acyclic class. Thus the component results compose, but operational equivalence prevents the stronger ontic inference. $\square$

Remark 1 (What is genuinely new). *The theorem does not claim a new component proof. Its content is the exact dependency statement: once temporalizability, actuality, stabilization and observer resolution are required on one typed packet, the claimed temporal story is coherent and no step may be removed or moved backward. In particular, an exact order cochain without EOCC is static structure; EOCC without observer resolution is not a measured clock; and resolved histories do not prove ontic traversal.*

4.1 Exact temporalizability versus mere acyclicity

Let D be the oriented edge–vertex incidence matrix of a connected finite occurrence complex and let ω be a prescribed edge increment. A scalar τ with

$$D\tau = \omega, \quad \omega_e > 0 \tag{21}$$

exists if and only if every signed cycle period vanishes,

$$\sum_{e \in \gamma} s_{\gamma e} \omega_e = 0 \quad \text{for every cycle } \gamma, \quad [\omega] = 0 \in H^1. \tag{22}$$

Acyclicity guarantees some topological order, not integrability of arbitrary prescribed increments. A diamond whose two directed paths have unequal total increment is the minimal counterexample. Conversely, positive exact increments forbid a directed active cycle. Equation (22) is therefore the safe global criterion; local closure can be insufficient on nontrivial topology.

4.2 Actuality is irreducible to a static action

The same static pair $(\mathcal{A}_{\text{occ}}, \Gamma)$ admits

$$X_- = -\Gamma^{-1}d\mathcal{A}_{\text{occ}}, \quad X_0 = 0, \quad X_+ = +\Gamma^{-1}d\mathcal{A}_{\text{occ}}. \tag{23}$$

All three have the same stationary body and action but opposite, absent or descending actuality. EOCC is precisely the enriched law retaining X_- . The response form follows conditionally from positive local Rayleigh minimization, in the Onsager variational spirit [12]; the fact that a response is physically actual does not follow from convexity alone. This is the weakest step of the entire temporal program and remains a Nature-level source-selection blocker.

Nor does ω_{occ} alone determine the full trajectory: any $Y \in \ker d\mathcal{A}_{\text{occ}}$ preserves its positive evaluation while moving within an action level set. The complete response law is needed to remove this solenoidal freedom inside the adopted model.

Temporalization, actuality and observer descent

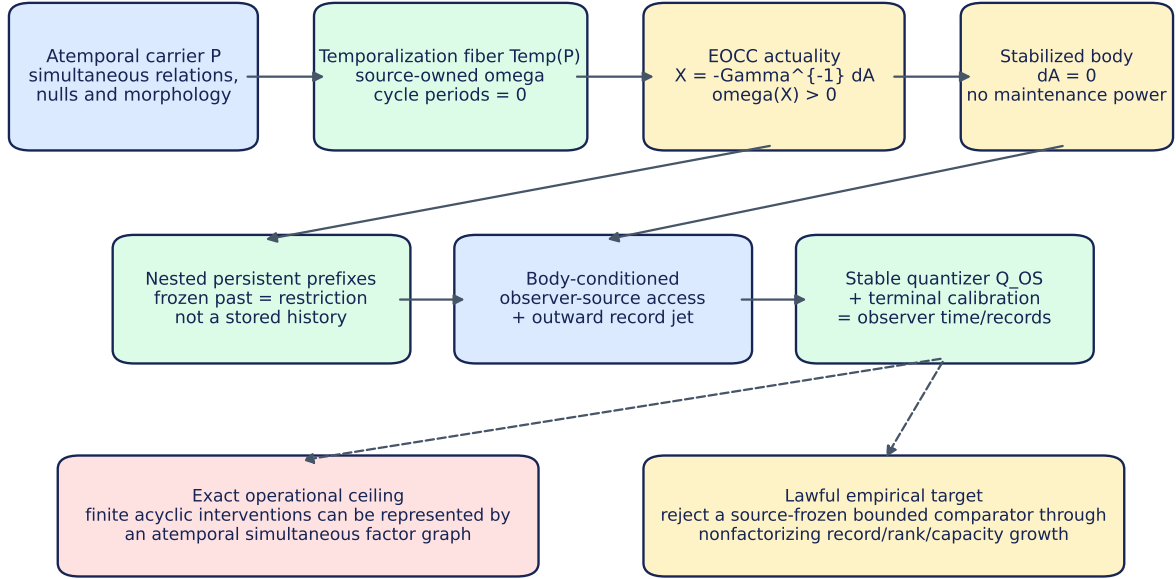


Figure 2: Temporalization and observer descent. The upper row separates an atemporal carrier, an integrable order structure, EOCC actuality and the stabilized endpoint. The middle row shows persistent prefixes and resolved observer records. The lower row records the operational ceiling and the only lawful empirical comparison.

5 Morphogenesis, progressive stabilization and the frozen past

5.1 The base executes a seed-owned relational program

The current synthesis assigns distinct roles:

seed: generative relational key,	base: susceptibility and response,	
body: realized stable response		(24)

A representative local response law is

$$\Gamma(M)\dot{M} = -D_M \mathcal{A}_{B,s_U}(M). \quad (25)$$

This does not place a material seed next to a distant front. The fixed seed acts through its representation in the already realized body and the base-owned local relation law. A formed prefix can therefore mediate access to the next admissible relation without the original seed being spatially adjacent to that front. The analogy to a signal traversing a network is only safe after one states which invariant the network preserves.

The relay result is asymmetric. A protected motif or relation class can be transported through lossy local maps and remain coherent by induction. But attenuation alone does not choose between smaller subsequent changes and longer delays. In a linear zero-threshold response it can reduce amplitude while leaving a normalized relaxation time unchanged. Genuine slowdown requires an additional source-owned mechanism such as cumulative body-additive drag, a moving minimum, differentiated response modes or a threshold. Pure edge-difference damping is an exact same-stiffness control that need not slow with depth.

5.2 Progressive hardening is weaker than exact freezing

Let P_a denote the realized prefix below progress level a . The coherent-extension law is

$$a \leq b \implies P_a \subseteq P_b, \quad \text{Res}_{ba}(M_b) = M_a. \quad (26)$$

This is the precise content of a body becoming rigid toward the past. It does not require a stored history. Earlier relations persist as restrictions of the later compatible body.

Exact finite-parameter freezing is nongeneric for smooth positive gradient flows; a regular solution often approaches equilibrium asymptotically. Operational freezing is weaker and sufficient. If a stable observer cell cannot distinguish $M(\sigma)$ from M_* , then a finite cell-entry parameter exists even when exact equality occurs only at infinity. In the linear scalar control

$$q(\sigma) - q_* = (q_0 - q_*)e^{-\sigma/\tau_{\text{rel}}}, \quad (27)$$

the resolved stabilization parameter scales as

$$\sigma_{OS}^{\text{stab}} = \tau_{\text{rel}} \log \frac{\|q_0 - q_*\|}{\delta_{OS}} + O(1), \quad (28)$$

where δ_{OS} abbreviates a source-owned stable-cell scale rather than defining readout equivalence by a nontransitive distance threshold. Thus a coarse observer can see a frozen body while a finer observer still resolves residual formation.

5.3 No passive maintenance power at the endpoint

Along EOCC,

$$\frac{d\mathcal{A}_{\text{occ}}}{d\sigma} = -\langle d\mathcal{A}_{\text{occ}}, \Gamma^{-1}d\mathcal{A}_{\text{occ}} \rangle. \quad (29)$$

If the body is a joint stable minimum of the base and seed, the endpoint has zero response current. The persistent seed is therefore best interpreted as a relational asymmetry defining the minimum, not a fuel packet consumed to hold the finished body in place. Action is exchanged or dissipated during formation and rearrangement; a static stable endpoint need not require continuous power. Removing or changing the seed may move the minimum and restart response, but that is a different intervention.

6 Observer time, resolved records and history refinement

6.1 A continuous precursor is not yet an operational tick

Let $\mathcal{R}_{OS}$ be the complete source-frozen record descriptor and let C_0 be the stable quantizer cell containing the zero precursor. On a local formation segment $d(\delta)$, suppose

$$\mathcal{R}_{OS}(d(\delta)) - \mathcal{R}_{OS}(0) = \delta a + \eta(\delta), \quad \|\eta(\delta)\| \leq \frac{C}{2}\delta^2. \quad (30)$$

If $B_{r_-} \subseteq C_0 \subseteq B_{r_+}$, then

$$\delta\|a\| + \frac{C}{2}\delta^2 < r_- \implies Q_{OS}(d(\delta)) = Q_{OS}(0), \quad (31)$$

$$\delta\|a\| - \frac{C}{2}\delta^2 > r_+ \implies Q_{OS}(d(\delta)) \neq Q_{OS}(0). \quad (32)$$

These Paper-VI results are imported to make one point: a nonzero Parent or channel derivative need not be a resolved observer transition. Time quantization is the smallest transition resolved by this observer–source instrument and context, not a universal atom of parent time.

On a constant-rank access stratum, only the outward Kato projector jet moves the accessible subspace. A complete exhaustive effect–Choi instrument detects every nonzero outward jet in at least one Choi record, while immediate effect probabilities and incomplete record subsets can remain blind. This is why an empirical temporal test must freeze the complete instrument rather than choose a favorable terminal after seeing the data. Kato perturbation theory supplies the standard mathematical control on the projector motion [13].

6.2 History refinement without a Parent history store

Let $(\mathcal{P}, \mu)$ be one source-frozen atemporal Parent ensemble and let X_a and D_b be simultaneous early and later record maps. The observer-relative compatible classes satisfy

$$\mathcal{H}_{x,d}^{(a,b)} = X_a^{-1}(x) \cap D_b^{-1}(d) \subseteq X_a^{-1}(x) = \mathcal{H}_x^{(a)}. \quad (33)$$

This is refinement by intersection, not later rewriting of an earlier Parent state. Stable record invariance alone is weaker than no-signalling. If the pre-choice weight μ is common to all later settings and the later instrument is represented by a normalized kernel $K_b(d|p)$, then

$$\sum_d P(x, d|a, b) = \int_{\mathcal{P}} \mathbf{1}_{X_a(p)=x} d\mu(p) = P(x|a), \quad (34)$$

independently of b . Conditioning on d may reveal different subensembles without creating a signal to the earlier observer. Delayed-choice and eraser-type statistics therefore fit Parent-fiber refinement but do not prove Tau Core.

6.3 Positive proto-order and negative weak times

A post-selected weak-time quantity may be negative even when the underlying actual response order is positive. The observer quantity depends on preparation, interference and postselection, schematically

$$T_{\text{weak}} = F(\text{interaction, preselection, postselection}), \quad (35)$$

whereas $\omega_{\text{occ}}(X_{\text{EOCC}}) > 0$ concerns the Parent formation packet. A finite compatibility witness exists because conditional quasi-averages need not be positive durations. This is consistent with weak measurement theory [14]; it is not evidence for a Tau-specific deviation or backwards Parent time.

6.4 Record persistence is not an entropy arrow

The occurrence action, observer order and entropy are different typed objects. For a differentiable parent lift $\mathcal{S}_P = \mathcal{S}_O \circ Q_{OS}$, EOCC gives

$$\frac{d\mathcal{S}_P}{d\sigma} = -\langle d\mathcal{S}_P, \Gamma^{-1} d\mathcal{A}_{\text{occ}} \rangle. \quad (36)$$

Consequently the exact local alignment condition for nondecreasing $\mathcal{S}_P$ is

$$\boxed{\langle d\mathcal{S}_P, \Gamma^{-1} d\mathcal{A}_{\text{occ}} \rangle \leq 0.} \quad (37)$$

Equation (37) is necessary and sufficient locally, but it is not implied by action descent. Two entropy functions with opposite gradients along the same EOCC orbit give opposite entropy arrows while preserving the action, occurrence order and stable-record packet. Thus the identification $\mathcal{S}_P = -\mathcal{A}_{\text{occ}} + \text{const}$ is a possible completion, not a thermodynamic derivation.

There is nevertheless an exact readout-level result. Let Y_n be the old resolved classes and suppose the one complete later readout has the source-frozen non-erasing form

$$Y_{n+1} = \coprod_{y \in Y_n} \{y\} \times F_y, \quad c_n(y, f) = y, \quad (38)$$

with a consistent ensemble

$$p_n(y) = \sum_{f \in F_y} p_{n+1}(y, f). \quad (39)$$

The Shannon chain rule then yields

$$\boxed{H(Y_{n+1}) - H(Y_n) = H(F_n | Y_n) \geq 0.} \quad (40)$$

The inequality is strict exactly when the new front effect resolves variation hidden inside at least one old class. If Q_n is the old quantizer and f_n the relative new effect, the exact finite criterion is

$$\exists x, x' : \quad Q_n(x) = Q_n(x'), \quad f_n(x) \neq f_n(x'). \quad (41)$$

On a regular smooth stratum this becomes

$$\text{rank} \begin{bmatrix} DQ_n \\ Df_n \end{bmatrix} > \text{rank } DQ_n, \quad \ker DQ_n \not\subseteq \ker Df_n. \quad (42)$$

These are imported net results of the morphogenesis program; their detailed finite controls remain with the canonical source. They prove accessible readout refinement, not creation of new Parent information. Indeed a finer observer can have increasing $H(Y_n)$ while its loss $H(X | Y_n)$ decreases. Neither quantity is automatically physical thermodynamic entropy. That promotion still requires a source-owned macroterminal, conserved quantities, an equilibrium measure and a standard thermodynamic limit; see the standard information-theoretic distinction in Cover and Thomas [15].

The corrected arrow statement is therefore conditional:

$$\boxed{\begin{aligned} &\text{odd source orientation} + \text{EOCC actuality} + \text{persistent records} \\ &+ \text{non-erasing visible refinement} + \text{consistent ensemble} + \text{thermodynamic identification} \\ &\implies \text{aligned actuality, record and thermodynamic arrows.} \end{aligned}} \quad (43)$$

The present source packet does not derive the last three physical premises or the low-entropy boundary. Hence this composition does not solve the Past Hypothesis; it isolates what a Tau-side solution would still have to own.

6.5 Common ownership no-go and minimal free-action completion

Even one source packet owning the occurrence law and the complete readout does not fix the entropy sign. At an occupied point in $X = \mathbb{R}^2$, choose

$$\mathcal{A} = x_1 + \frac{1}{2}x_2^2, \quad Q = x_1, \quad f = x_2, \quad \Gamma = I. \quad (44)$$

Then $X_{\text{EOCC}} = -e_1$, while $\ker DQ = \text{span}(e_2)$ and $Df(e_2) = 1$, so the same packet has strict hidden-fiber front novelty. Nevertheless the two scalar terminals $\mathcal{S}_+ = -x_1$ and $\mathcal{S}_- = x_1$ give

$$d\mathcal{S}_+(X_{\text{EOCC}}) = +1, \quad d\mathcal{S}_-(X_{\text{EOCC}}) = -1. \quad (45)$$

Thus common ownership, EOCC and front nonfactorization do not entail a thermodynamic arrow.

A sufficient completion is source-owned free-action structure

$$\mathcal{A}_{\text{occ}} = \mathcal{E}_P - \Theta_P \mathcal{S}_{\text{th}}, \quad \Theta_P > 0, \quad d\mathcal{E}_P(X_{\text{EOCC}}) = d\Theta_P(X_{\text{EOCC}}) = 0. \quad (46)$$

Combining Eq. (46) with EOCC yields

$$\boxed{\frac{d\mathcal{S}_{\text{th}}}{d\sigma} = \frac{1}{\Theta_P} \langle d\mathcal{A}_{\text{occ}}, \Gamma^{-1} d\mathcal{A}_{\text{occ}} \rangle \geq 0,} \quad (47)$$

strictly away from equilibrium. Conservation is essential: if $d\mathcal{E}_P(X_{\text{EOCC}})$ is sufficiently negative, the entropy rate reverses while EOCC still descends. The discrete front result becomes thermodynamic under the stronger optional macroterminal identity

$$\mathcal{S}_{\text{th},n+1} - \mathcal{S}_{\text{th},n} = k_B H(F_n | Y_n) + \Sigma_n, \quad \Sigma_n \geq 0. \quad (48)$$

Equations (46) and (48) are additional enriched source laws, not consequences of common source naming. The latter is sufficient, not necessary. Current Tau has not derived their physical ownership, the equilibrium measure defining $\mathcal{S}_{\text{th}}$, or the initial low-entropy boundary.

6.6 Configuration-Laplacian free action and the front-entropy no-go

The conserved-macrofiber program supplies a weaker and more natural positive route. On a connected finite configuration carrier $\Sigma_m = \{1, \dots, N\}$, let

$$(Lp)_i = \sum_j w_{ij}(p_i - p_j), \quad w_{ij} = w_{ji} \geq 0, \quad \text{quad}_i = N^{-1}. \quad (49)$$

For positive p , define the logarithmic mean

$$\Lambda(a, b) = \frac{a - b}{\log a - \log b}, \quad \Lambda(a, a) = a, \quad (50)$$

and the positive Onsager operator

$$(\mathcal{K}_p \xi)_i = \sum_j w_{ij} \Lambda(p_i, p_j) (\xi_i - \xi_j). \quad (51)$$

With

$$\mathcal{A}_{\text{th}}(p) = \vartheta D(p \| u), \quad \Gamma_p^{-1} = \vartheta^{-1} \mathcal{K}_p, \quad (52)$$

the identity $\Lambda(p_i, p_j)(\log p_i - \log p_j) = p_i - p_j$ gives

$$\boxed{\dot{p} = -\Gamma_p^{-1} d\mathcal{A}_{\text{th}} = -Lp.} \quad (53)$$

Thus symmetric macrofiber mixing is exactly an EOCC gradient flow under the source-derived logarithmic-mean mobility. Setting

$$\mathcal{S}_{\text{th}} = k_B H(p), \quad T_P = \vartheta/k_B, \quad \mathcal{E}_m = \vartheta \log N, \quad (54)$$

yields

$$\boxed{\mathcal{A}_{\text{th}} = \mathcal{E}_m - T_P \mathcal{S}_{\text{th}}} \quad (55)$$

and

$$\boxed{\frac{d\mathcal{S}_{\text{th}}}{d\sigma} = \frac{k_B}{2} \sum_{i,j} w_{ij} (p_i - p_j) (\log p_i - \log p_j) \geq 0.} \quad (56)$$

Equality holds only at the uniform state on a connected carrier. This is the infinitesimal form of the conditional macrofiber H-theorem already owned by the foundation program.

The global occurrence action can be restricted to $\mathcal{A}_{\text{th}}$ only if one physical incidence identifies the occupied temporal front with this same conserved configuration carrier and its positive activation. The present configuration-graph and convex-weight packets make that construction nonempty, but do not prove the identification.

Nor does Eq. (48) follow. At equilibrium on a uniform two-state carrier, thermal mixing gives $\Delta \mathcal{S}_{\text{th}} = 0$. A constant old readout and a new front label that distinguishes the two states gives $H(F | Y) = \log 2 > 0$. The carrier and measure can be common, yet the proposed identity would require $\Sigma = -k_B \log 2 < 0$. Therefore accessible front refinement and thermodynamic production

may share a source and orientation without having equal increments. The remaining physical target is one front-to- configuration incidence, not FMT-P1.

That target is not already implied by the two existing packets. Keep fixed the complete seed-base response reduct and a two-state symmetric thermal carrier with positive activation. One extension identifies the two front labels bijectively with the two thermal configurations. A second extension keeps every upstream object fixed but admits only the constant incidence from both front labels to one thermal state. A complete two-label front readout cannot factor through that rank-one incidence. Hence

$$\boxed{\text{SBMR} + \text{configuration heat} \not\Rightarrow \text{front- configuration incidence.}} \quad (57)$$

The conclusion remains true after positive thermal activation is supplied; the missing datum is cross-typed registration rather than a rate.

A minimal positive completion is exact. Let one primitive source alphabet $\mathcal{A}_{\text{src}}$ carry source-owned bijections

$$e_F : \mathcal{A}_{\text{src}} \xrightarrow{\sim} F, \quad e_\Sigma : \mathcal{A}_{\text{src}} \xrightarrow{\sim} \Sigma_m, \quad \chi = e_\Sigma e_F^{-1}. \quad (58)$$

If their pullbacks agree on the complete macroterminal, measure and local rewrite incidence, χ is the unique incidence with those properties. Positive source depth u additionally gives $\lambda(u) = 2u/\ell_* > 0$ under the existing positive-mode depth calibration. This proves the common-carrier arrow alignment conditionally. The current Tau source inventory does not derive the primitive atomic co-ownership in Eq. (58); equal cardinality or common source naming is insufficient.

Nor does one universal seed force that premise. Grant the one structured seed the split algebra

$$\mathcal{A}_U = \mathbb{R}p_0 \oplus \mathbb{R}p_1, \quad p_a p_b = \delta_{ab} p_a, \quad p_0 + p_1 = 1. \quad (59)$$

Keep the front action faithful, $\rho_F(p_0) = \text{diag}(1, 0)$ and $\rho_F(p_1) = \text{diag}(0, 1)$. The same one-seed reduct admits both the matching faithful thermal action $\rho_\Sigma^+ = \rho_F$ and the unital nonfaithful action

$$\rho_\Sigma^-(p_0) = I_2, \quad \rho_\Sigma^-(p_1) = 0. \quad (60)$$

Both may carry the same two-state thermal Laplacian, measure and positive activation. The second action collapses one primitive seed atom and cannot support atomic co-ownership. Thus universal-seed uniqueness is insufficient even after a primitive algebra is supplied.

The corrected sufficient theorem is finite. If a split algebra with N primitive idempotents acts faithfully, unitaly and multiplicity-freely on both V_F and V_Σ , with $\dim V_F = \dim V_\Sigma = N$, then its pairwise orthogonal nonzero primitive projectors have positive integer ranks summing to N . Every rank is therefore one. The common primitive spectrum consequently fixes exactly one front atom and one thermal atom per idempotent, and hence fixes their incidence up to simultaneous algebra relabeling. If the measure, macroterminal and rewrite graph also live on that primitive spectrum, Eq. (58) and the conditional arrow alignment follow. The existing PUSC structure is a candidate primitive algebra, but the two physical faithful module actions and common primitive rewrite ownership are not yet derived.

These representation premises admit a sharper source-side compression. Let $\mathcal{A}_U \cong \mathbb{R}^N$ act unitaly on an N -dimensional carrier V_X , and let a source-descended root v_X define

$$C_X : \mathcal{A}_U \longrightarrow V_X, \quad C_X(a) = \rho_X(a)v_X. \quad (61)$$

If v_X is cyclic, C_X is onto and hence an isomorphism. Therefore ρ_X is faithful. Moreover the vectors $C_X(p_a)$ form a basis and $\text{im } \rho_X(p_a) = \mathbb{R}C_X(p_a)$, so every primitive atom occurs exactly once. Applied separately to the front and thermal carriers, paired cyclicity replaces faithfulness, multiplicity freedom and atomic exhaustivity by the single certificate

$$\gamma_{\text{pair}} = \min_{X \in \{F, \Sigma\}} \lambda_{\min}(C_X^* C_X) > 0. \quad (62)$$

Together with the common primitive measure, macroterminal and rewrite graph, this condition implies the unique incidence in Eq. (58). It is not implied by common ancestry. In the control of Eq. (60), choose $v_F = (1, 1)^T$ and $v_\Sigma = (1, 0)^T$. Then $C_F = I_2$ but $C_\Sigma = \text{diag}(1, 0)$, so the same seed algebra has a cyclic front root and a singular thermal root. The current protected-parent cyclicity result therefore supplies a template, not a proof that both post-body descents preserve cyclicity.

The primitive joint Hessian now makes the source origin of this condition explicit. After body stabilization, let E_U carry a positive source metric K_U , let $H_X > 0$ be the front or thermal response Hessian, and let B_X be the mixed seed-carrier block. The quadratic two-jet

$$\mathcal{A}_X(a, z_X) = \frac{1}{2} z_X^* H_X z_X - z_X^* B_X a + \frac{1}{2} a^* K_U a \quad (63)$$

gives, by stationarity,

$$C_X = H_X^{-1} B_X, \quad W_X = C_X^* H_X C_X = B_X^* H_X^{-1} B_X. \quad (64)$$

Positivity of H_X implies $\ker W_X = \ker B_X = \ker C_X$. Therefore paired cyclicity is equivalent to

$$\ker B_F = \ker B_\Sigma = \{0\}. \quad (65)$$

The metric form $W_X \succeq \epsilon_X K_U$ with $\epsilon_X > 0$ gives the explicit bound

$$\gamma_{\text{pair}} \geq \min_X \frac{\epsilon_X \lambda_{\min}(K_U)}{\lambda_{\max}(H_X)} > 0. \quad (66)$$

Equivalently, if one full-rank master coupling B_U is followed by $B_X = T_X B_U$, it is sufficient that $\ker T_X \cap \text{im } B_U = \{0\}$ on both legs. This is protected non-erasure, not global losslessness.

Neither stability nor combined completeness forces this result. Taking $K_U = H_F = H_\Sigma = I_2$, $B_F = I_2$ and $B_\Sigma = \text{diag}(1, 0)$ leaves both carrier Hessians positive and the stacked coupling full rank, while the thermal work Gram is singular. Thus a stable thermal response may remain blind to one seed atom. The exact common-unit identity $W_F = W_\Sigma = K_U$ is a sufficient source/root Gram-isometry completion, but the current Tau action does not derive it for these two typed carriers.

The zero-kernel requirement is stronger than necessary for a lossy Tau descent. Put $K_F = \ker B_F$ and $K_\Sigma = \ker B_\Sigma$. A quotient E_U/K admits well-defined and faithful induced maps on both carriers if and only if

$$K_F = K_\Sigma =: K_{\text{com}}. \quad (67)$$

Indeed, descent requires $K \subseteq K_X$, whereas faithfulness requires $K_X \subseteq K$. Thus E_U/K_{com} is the unique maximal common faithful source quotient. A zero common kernel gives complete universal-seed alignment. A nonzero common kernel gives only quotient-level alignment unless the erased sector is proved null for every claimed macroterminal.

Equal ranks do not suffice. The pair $B_F = \text{diag}(1, 0)$, $B_\Sigma = \text{diag}(2, 0)$ has one faithful common quotient, while $B_F = \text{diag}(1, 0)$, $B_\Sigma = \text{diag}(0, 1)$ has equal ranks but different kernels. Both can descend from the same injective master body coupling. Consequently, even if every seed atom is essential to formation of the complete body, lossy post-body descents need not erase the same atoms. The least physical completion is common protected-kernel descent, not universal losslessness.

7 Operational equivalence and the limit of finite experiments

7.1 Finite interventional simulation no-go

Let $G = (V, E)$ be a finite acyclic graph with local response laws

$$x_v = f_v^{i_v}(x_{\text{Pa}(v)}, \xi_v), \quad (68)$$

including predeclared interventions, adaptive decision nodes and shared exogenous variables. The simultaneous relation

$$\mathcal{S}_i = \{(x, \xi) : x_v - f_v^{i_v}(x_{\text{Pa}(v)}, \xi_v) = 0 \ \forall v\} \quad (69)$$

has the same deterministic solutions. For stochastic kernels, the product

$$\mu_i(dx) = \prod_{v \in V} K_v^{i_v}(dx_v \mid x_{\text{Pa}(v)}) \quad (70)$$

is both the sequential sampling law and the atemporal factor-graph law. Shared exogenous variables preserve the counterfactual coupling. Hence every terminal and joint table in this class agrees.

The result resembles standard equivalence issues in causal modeling [16], but its role here is ontological discipline: manipulating an internal process does not prove that the Parent literally traverses the algorithm used to generate the same table. The simultaneous representation is not a memory store or an instruction to run a solver in hidden time.

7.2 What can still be tested

The no-go does not make the temporal completion scientifically empty. It changes the admissible null hypothesis. A decisive class-relative test must freeze before endpoint inspection:

- (D1) a source-complete bounded atemporal comparator $\mathcal{C}_A$;
- (D2) a temporal completion $\mathcal{C}_T$ with no post-result source or access enlargement;
- (D3) one common intervention cut and complete observer instrument;
- (D4) nuisance, wrong-family and contextual-access controls;
- (D5) an observable whose prediction sets are disjoint or whose complete response rank exceeds the frozen operational capacity of $\mathcal{C}_A$.

One local linear form is

$$\Delta r_k = \text{rank} \begin{pmatrix} J_{\text{old}} \\ J_k \end{pmatrix} - \text{rank}(J_{\text{old}}), \quad (71)$$

after quotienting standard nuisance tangents and freezing the common cut. Positive prefix rank growth rejects only a comparator whose complete effect module has been bounded independently. An unrestricted contextual atemporal descriptor can add a separate context sector and saturate any finite table. Calling that unrestricted class “falsified” would merely punish it for not being the bounded model that was tested.

Proposition 1 (Empirical proof ceiling). *Finite terminal agreement or disagreement can select between predeclared bounded model classes. It cannot by itself prove literal Parent traversal against the unrestricted atemporal class containing the projected temporal completion. Nature-level temporal evidence requires both a source-derived comparator-capacity restriction and a terminal effect that survives the full observer/source/nuisance quotient.*

This proposition is a synthesis of the finite saturation, common-cut and source-cost results. It prevents the atemporal model from becoming an unfalsifiable “anything fits” explanation while also preventing an artificially weak comparator from manufacturing temporal evidence.

8 Formation budget, four-dimensional exchange and dark-sector limits

8.1 Finite parent-action budget

For a regular EOCC section ending at a stable minimum,

$$\Delta\mathcal{A}_{\text{form}} = \mathcal{A}_{\text{occ}}(q_0) - \mathcal{A}_{\text{occ}}(q_*) = \int d\sigma \langle d\mathcal{A}_{\text{occ}}, \Gamma^{-1} d\mathcal{A}_{\text{occ}} \rangle. \quad (72)$$

This is an exact parent-action identity inside the working completion. It is not yet energy in joules, a local density or stress-energy. The conversion requires a terminal metric, time direction, field normalization and action unit. These belong downstream and may not be chosen from a dark-sector fit.

8.2 One exact conditional 4D soldering

As a nonempty terminal completion, introduce positive scales

$$t = t_*\sigma, \quad \varphi = \varphi_*q, \quad V(\varphi) = \varepsilon_*\mathcal{A}_{\text{occ}}(q), \quad (73)$$

and a future timelike terminal vector u^μ . The first-order law

$$\Upsilon D_t \varphi + V_{,\varphi} = 0 \quad (74)$$

can reproduce the scalar EOCC orbit for a corresponding positive Υ . The algebraic form stress

$$T_{\mu\nu}^{\text{form}} = -V g_{\mu\nu} \quad (75)$$

is not separately conserved during formation:

$$\nabla_\mu T_{\text{form}}^{\mu\nu} = J_{\text{form}}^\nu = -V_{,\varphi} \nabla^\nu \varphi. \quad (76)$$

Bianchi compatibility requires a receiver with

$$\nabla_\mu T_{\text{res}}^{\mu\nu} = -J_{\text{form}}^\nu. \quad (77)$$

The integrated transfer equals $\varepsilon_*\Delta\mathcal{A}_{\text{form}}$ under this soldering, but the three conversion scales remain free upstream.

8.3 Why this does not derive dark matter or dark energy

The same homogeneous positive source $S(t) = \Upsilon\dot{\varphi}^2$ admits receiver equations

$$\dot{\rho}_w + 3H(1+w)\rho_w = S(t) \quad (78)$$

for $w = -1, 0, 1/3$, giving vacuum-like, dust-like and radiation-like backgrounds. Conservation therefore fixes loading but not pressure, sound speed, anisotropic stress, localization, clustering or the split among receivers. The scalar formation packet supplies no unique halo, lensing or cosmological-constant prediction.

Moreover, an atemporal completed-orbit model with the same restricted orbit and terminal soldering reproduces $(\varphi, T_{\mu\nu}^{\text{form}}, J_{\text{form}}^\nu)$. The conditional 4D exchange current is therefore not an actuality-nonfactorizing discriminator. The robust conclusion is narrower:

Temporal formation can conditionally source a finite 4D exchange packet, but neither its observed existence nor its total budget proves Parent temporality or selects a dark sector.

Family	What was obtained	Current temporal verdict
Resolved galaxy morphology/age	Frozen clock-blind geometry, null or ordinary-evolution-compatible endpoints, source-label and covariance audits	No formation-specific temporal signal; later terminal selection forbidden
Atomic and many-body ramps Glass/Kovacs data	Stage-resolved or return-path method preflights History-dependent public response with present-state normalization controls	Useful controls, but no Parent bridge and no temporal discrimination Public packet does not realize an identical complete present-state comparison
Photonic circuits	Executable design for joint onset/coherence method validation; partial public resources	Hardware acquisition packet, not Tau temporality evidence
Quantum process/tomography tables	One centered rank-excess preflight survived several simple nuisance projections	Broader operational cut and cross-run replication fail; aggregate drift remains non-identifiable
Negative weak-time experiments	Operational example of conditioned time not being a positive classical duration	Compatible with positive Parent order; fully explained within standard quantum theory
JWST/Hubble/CMB targets	Reparameterization and sign audits	A universal one-lapse deformation has no independent content or wrong joint sign; no Tau signal established

Table 2: Preserved net status of the observational program. “Preflight” is not empirical validation. The source documents retain dataset provenance and negative controls.

9 Observational program and preserved negative results

The temporal program inspected several public-data families. None currently crosses the source-to-terminal and comparator-capacity gates. Table 2 records the net status rather than retelling every preflight.

The strongest provisional process-tomography feature was a structured centered rank excess in one released aggregate packet. It did not replicate across the wider public run family, and the released aggregates do not identify the missing drift or context bound. Preserving this demotion is scientifically more valuable than treating every unexplained residual as temporal evidence.

For cosmology, a mere reparameterization $t_{\text{obs}} = F(\sigma)$ is physically empty unless the map changes a source-owned relation between independently measurable sectors. A universal geometric lapse that gives early systems more parent formation opportunity does not automatically have the sign required to relieve both early-structure and late-distance tensions. JWST, Hubble and CMB results may motivate a future sector-specific readout law, but they do not currently select it.

10 Proof ladder, falsification and current status

The complete program is best summarized as a ladder.

L0: definitions

Atemporal carrier, temporalization fiber, seed key, observer/source quantizer and terminal maps.

L1: exact finite mathematics

Cycle-period temporalizability, action cochain exactness, positive EOCC contraction, stable-cell window, complete record detection and no-retro marginalization.

L2: conditional mechanism

One enriched seed–base/occurrence action, positive response form, EOCC actuality, progressive stabilization and finite formation budget.

L3: physical source selection

Nature-level ownership and occupation of the complete action, actuality law, front/access co-registration, instrument and calibration. This level is open.

L4: empirical discrimination

Source-frozen bounded comparator, nonfactorizing complete record and independent replication. This level is open.

The strongest current result is L2. It is stronger than a verbal analogy: the conditional response, order and budget are explicit and internally audited. It is weaker than a physical theory of time because L3 is an additional premise and L4 has no successful endpoint.

10.1 Failure and demotion conditions

The temporal working completion would be weakened or rejected in its declared domain if any of the following occurs:

- (F1) the source-owned occurrence cochain has a nonzero period after the true physical-null quotient while the model claims one global scalar proto-time;
- (F2) the proposed response metric is not positive on the retained physical quotient or the EOCC section fails the claimed action descent;
- (F3) the prefix law overwrites a previously stable source-owned record;
- (F4) the claimed observer clock depends on a nontransitive threshold rather than a stable quantizer partition;
- (F5) a purported temporal signal vanishes under the complete source-owned instrument, wrong-family or contextual-access controls;
- (F6) a Parent front rate is inferred from one observer tick series while the channel transmission or stable-cell density is fitted from that same series rather than frozen independently;
- (F7) the same signal is reproduced by the predeclared bounded atemporal comparator without exceeding its frozen capacity;
- (F8) a 4D energy or dark-sector claim requires an equation of state, localization or unit scale not derived upstream.

Conversely, a serious strengthening would require a single source-owned packet that simultaneously fixes the action cochain, actuality law, formation-front/access first jet, complete record instrument and stable quantizer, followed by an independently frozen observation exceeding a predeclared bounded atemporal comparator. No currently registered packet closes that chain.

11 Discussion

The original intuition—a morphological body rigid toward the past and fluid toward the future—survives in a corrected form. “Rigid” means restriction-compatible persistence of actualized prefixes and records. “Fluid” means the current temporalization admits more than one unresolved or not-yet-actualized continuation at the observer’s resolution. It does not mean that the atemporal carrier lacks all future relations, nor that a stored past grows inside the Parent.

The seed/base analogy also survives with stricter typing. The seed supplies a fixed relational grammar in the language to which the base responds. The base supplies stiffness, susceptibility, mediation and damping structure. Their joint law defines a stable body. A temporal completion adds the actuality of ordered response; it does not transform the seed into a material energy capsule. Instability is admissible, but our realized stable universe selects a stable occupied branch only conditionally.

The deepest conceptual gain is the separation of four notions often called “time”:

$$\boxed{\text{orientation/order} \neq \text{actual response parameter} \neq \text{observer clock} \neq \text{SI calibration.}} \quad (79)$$

Tau Core can consistently place these at different layers. This removes the need to put ordinary 4D time directly into the Parent, but it also prevents the theory from claiming more than it has derived.

The entropy audit adds a parallel separation. EOCC action descent fixes an actuality orientation inside the adopted completion; persistent records fix a record arrow; visible non-erasing refinement fixes an accessible-information arrow. A thermodynamic arrow follows only after a physical macroterminal and its statistical-mechanical measure are derived. Common source ownership does not close this gap: the exact counterfamily permits opposite entropy arrows on the same EOCC/readout/front reduct. The free-action identity closes the local sign only conditionally on conserved Parent energy and a positive fixed conversion scale. On a conserved configuration carrier the symmetric Laplacian and logarithmic-mean mobility derive that free-action structure exactly; the remaining issue is physical front-to-configuration registration. The refined-front increment and thermodynamic production are distinct and need not be equal. A same-reduct countermodel proves that the SBMR and heat packets do not force that registration even with positive activation. The unique incidence follows only under primitive atomic co-ownership of the front and thermal presentations, which remains physically unselected. One universal seed does not remove this premise: the same split seed algebra admits faithful and nonfaithful thermal actions. Faithful unital multiplicity-free actions with dimension equal to the primitive count do force rank-one atomic matching, but their physical ownership remains open. Equivalently, a cyclic source root on each dimension-matched carrier forces all three properties, and the exact paired Gram condition in Eq. (62) tests them without terminal entropy data. The current action does not yet derive that both post-body source evaluations are nonsingular. Observer existence alone does not choose that arrow, and record information may grow while observer loss decreases.

The operational no-go is equally important. If an unrestricted atemporal model is allowed to encode an independent context for every finite experiment, it has unlimited explanatory capacity and loses discriminatory value. The scientifically viable comparison is therefore not “all atemporal structures versus one temporal structure.” It is a frozen, physically sourced atemporal subclass versus a frozen temporal completion, both charged for their source-owned operational capacity. This is the route by which the temporally richer but narrower model can become more predictive without being declared true by construction.

12 Conclusion

The Tau Core temporal program now has a coherent architecture and a precise epistemic boundary. The parent remains atemporal at the carrier level. Temporalizable carriers admit additional source-owned fibers. A complete occurrence action gives an exact order cochain, and EOCC supplies positive actuality inside one explicitly adopted enriched completion. The resulting formation can stabilize progressively, preserve a frozen prefix, terminate without continuous maintenance power and descend to observer-relative clocks and records only through the full source-owned channel and quantizer.

What has not been proved is equally definite. Static Tau does not force actuality; the proto-order does not fix an absolute clock or SI unit; finite acyclic intervention tables do not identify

literal traversal; a formation budget is not automatically 4D energy; EOCC and stable records do not by themselves imply thermodynamic entropy growth or a low-entropy boundary; and no current observation selects the temporal completion or a dark sector. The next scientific task is not another internal reformulation. It is physical derivation or rejection of paired cyclic descent of the universal source root into the EOCC-front and conserved thermodynamic carriers, together with their common primitive graph; positive activation then follows from positive source depth under the declared depth calibration, together with one independently frozen, nonfactorizing empirical test against a bounded atemporal comparator. Any rejection from that test is model-class relative: neither a reduct-derived cost nor complete-cut rank distinguishes the temporal completion from its saturated image.

A Compact theorem and no-go ledger

Table 3: Net temporalization claims used by this manuscript.

Identifier	Statement	Status
TEMP-FIB	Temporal completions are fibers over temporalizable atemporal carriers	Typed architecture theorem
EXACT-TIME	A finite scalar proto-time exists iff signed cycle periods vanish after null descent	Proved
ACT-OMEGA	A complete quotient-basic action selects $\omega_{\text{occ}} = -d\mathcal{A}_{\text{occ}}$	Proved inside the declared source class
EOCC-ACT	Positive EOCC mobility makes $\omega_{\text{occ}}(X_{\text{EOCC}}) > 0$ off equilibrium	Proved inside the adopted actuality law
STATIC-NO	A static action and endpoint do not select descending, zero or ascending actuality	Exact counterfamily
PROG-STAB	EOCC relaxation gives monotone action descent and resolution-relative freezing	Conditional theorem
RELAY-NO	Loss alone selects neither longer latency nor an absolute clock scale	Exact non-entailment
OBS-TIME	Operational time needs access, typed clock precursor, stable quantizer and calibration	Conditional descent
OBS-RATE-ID	Resolved tick intensity factorizes into observer-cell density, channel transmission and Parent front rate; any number of uncalibrated clocks retains two scale gauges, and absolute rate also needs a Parent scale	Exact conditional non-identifiability
OBS-CAL	A jointly source-owned density, transmission and positive scale one-form define an invariant rate, but the calibrated record is also compatible with an atemporal ordered carrier	Conditional completion plus exact non-entailment
EOCC-CAL	The clock-mobility product $\hat{\Gamma} = \kappa_P \Gamma_\rho$ gives an invariant EOCC proto-time law and release rate	Proved conditionally
CAL-ORBIT-NO	The calibrated autonomous EOCC flow identifies literal traversal	Disproved by exact simultaneous-orbit representation

Identifier	Statement	Status
SAT-BAYES-NO	A calibrated temporal completion is observationally preferred to its saturated atemporal image	Disproved pairwise: exact Bayes factor one
NONFACT-REL	A body-orbit-nonfactorizing terminal excludes every atemporal ontology	Disproved; it tests only a predeclared bounded comparator
EXT-SRC-NO	A source law in the complete shared extensional Parent signature can exclude the saturated image	Disproved by identity of the reducts
CAP-SAT-NO	A capacity functional of the shared reduct or a complete-cut rank bound excludes only the saturated atemporal image	Disproved; both members have equal cost and paired cut rank
ENTROPY-NO	EOCC action descent, observer order and persistent records do not select an entropy arrow	Exact non-entailment
ENT-ALIGN	Local entropy growth along EOCC is equivalent to $\langle d\mathcal{S}_P, \Gamma^{-1}d\mathcal{A}_{\text{occ}} \rangle \leq 0$	Proved local criterion
READ-ENT	Non-erasing visible refinement with a consistent ensemble gives $\Delta H = H(F_n Y_n) \geq 0$	Proved conditionally
FRONT-NOVEL	Strict readout-entropy growth requires a nonfactorizing relative front effect inside an old readout fiber	Exact finite criterion
THERM-NO	Accessible readout entropy or observer-loss entropy is already thermodynamic entropy	Not proved
COMMON-SRC-NO	Common ownership of EOCC, the old record and a nonfactorizing front effect fixes the thermodynamic entropy sign	Exact counterfamily
FREE-ACT	A source-owned $\mathcal{A}_{\text{occ}} = \mathcal{E}_P - \Theta_P \mathcal{S}_{\text{th}}$, conserved $\mathcal{E}_P$ and fixed $\Theta_P > 0$ imply $d\mathcal{S}_{\text{th}}/d\sigma \geq 0$	Proved conditionally
LAP-FREE	A symmetric connected configuration Laplacian has an exact logarithmic-mean EOCC form, free-action decomposition and nonnegative thermodynamic entropy production	Proved on the declared macrofiber
FMT-NO	Common carrier, thermal mixing and visible front refinement force $\Delta \mathcal{S}_{\text{th}} = k_B H(F Y) + \Sigma$ with $\Sigma \geq 0$	Exact equilibrium countermodel
TFCI-NO	SBMR plus the configuration-heat packet forces physical front–configuration registration, even with positive activation	Exact same-reduct countermodel
TFCI-COND	Faithful front and thermal presentations of one primitive source alphabet fix the unique measure-, macroterminal- and incidence-preserving map	Proved conditionally
USAC-NO	One universal seed, even with one fixed split primitive algebra, forces faithful front and thermal atomic actions	Exact same-seed countermodel

Identifier	Statement	Status
USAC-COND	Faithful unital multiplicity-free N -dimensional actions of an N -atom source algebra force rank-one atomic matching	Proved conditionally
USCYCL-COND	A cyclic source root on each dimension-matched front and thermal module forces faithful exhaustive rank-one actions; one-sided cyclicity is insufficient	Proved conditionally plus exact same-seed control
PAIR-WORK	Primitive mixed blocks give $C_X = H_X^{-1}B_X$; paired cyclicity is equivalent to individual injectivity, while stability and stacked rank are insufficient	Proved conditionally plus exact stable rank-loss control
COM-KERNEL	Faithful common lossy descent exists iff the front and thermal mixed blocks have equal kernels; seed minimality does not force equality	Proved plus exact equal-rank countermodel
INT-SIM-NO	Finite source-frozen acyclic processes admit simultaneous atemporal factorization	Proved no-go
BOUND-DISC	Rank/capacity growth can reject a source-frozen bounded comparator	Conditional discriminator
SATURATION-NO	Unrestricted contextual atemporal classes saturate finite terminal tables	Proved ceiling
FORM-BUDGET	Integrated EOCC dissipation equals a finite parent-action drop	Proved inside the working completion
4D-EXCHANGE	One downstream soldering yields a conserved total formation–receiver exchange packet	Conditional construction
DARK-NO	Loading does not select receiver equation of state, localization or dark split	Proved non-entailment
CELL-WINDOW	A continuous record jet is resolved only after exiting a source-owned stable cell	Conditional exact bounds
CHOI-DETECT	An exhaustive complete Choi instrument detects every nonzero outward projector jet	Proved on the declared stratum
HISTORY-REFINE	Later conditioning intersects compatible Parent fibers without rewriting stable earlier records	Proved by set inclusion
NO-RETRO	Common preparation plus normalized future-local instrument preserves the earlier marginal	Proved conditionally
WEAK-TIME	Positive Parent order is compatible with negative post-selected weak-time values	Finite compatibility witness
NATURE-TEMP	Nature occupies the enriched temporal completion	Not proved

B Reproducibility controls

The public script checks thirteen finite statements:

1. a consistent weighted diamond has zero cycle period while an acyclic inconsistent diamond does not;
2. a positive quadratic action and response metric give strict EOCC action descent and a

- positive actuality certificate;
3. scalar relaxation has the expected finite formation-budget identity;
 4. a finite intervened DAG and its simultaneous relation have the same outputs for every tested intervention;
 5. a common preparation with normalized later kernels has a setting-independent earlier marginal.
 6. one EOCC orbit admits opposite entropy-rate signs without alignment, while a non-erasing visible refinement satisfies the Shannon chain rule and the hidden-fiber nonfactorization control.
 7. one common EOCC/readout/front packet admits opposite entropy signs, while the free-action completion gives positive entropy production and the energy-drift control reverses it.
 8. a logarithmic-mean Onsager operator reproduces symmetric Laplacian mixing and its free-action entropy production, while the equilibrium refined-front control rejects generic FMT-P1.
 9. the same SBMR/thermal reduct admits both bijective and rank-one constant front incidences, while a common primitive source alphabet gives the unique bijective lift and positive depth gives positive activation.
 10. one fixed two-atom universal-seed algebra admits faithful and unital nonfaithful thermal actions, while faithful exhaustive actions recover the unique rank-one common-atom matching.
 11. cyclic front and thermal source-evaluation maps have positive Gram spectra, while the same seed algebra admits an invertible front map and a singular noncyclic thermal control.
 12. primitive mixed Hessian blocks reproduce the source-work Grams; full-rank blocks are positive, while a positive stable thermal Hessian and a full-rank stacked coupling can retain a singular thermal leg.
 13. equal singular kernels yield faithful induced maps on a common quotient, while equal-rank unequal-kernel descents arise from the same injective master body coupling and admit no faithful common quotient.

These are sanity checks of the displayed finite constructions, not substitutes for the analytic proofs and not empirical evidence.

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